RHYTHM PLUS FIRMWARE A0120 · CIRCUIT RHYTHM

Pattern Mutate and Track +

New ways to vary your patterns and to shape each track. This chapter shows every control, page and light.

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> The drawings show the pads and buttons of the Circuit Rhythm as a plan, not to scale. The colours are close to the colours on the unit; on the unit, a dim pad is darker than it is on paper.

CHAPTER 1

Pattern Mutate: overview and quick start

A mutate gives each track a new play list: a short run of that track's own patterns, in a random order. The track plays the list in a loop in place of its chain. Your patterns do not change. You can go back to the chains at any time.

Use it to find new variations of a groove: try a mutate, keep it if you like it, try another if you do not.

Quick start

1. Press **Patterns** to show the Patterns page. Press **Play**.
2. Hold **Shift** and tap **Duplicate**. Each track gets a new list. The lists start at each track's next pattern end.
3. Tap again for another mutate. Each tap makes new lists.
4. To go back to the chains: hold **Clear** and **Shift**, and tap **Duplicate**.

The Duplicate button

On the Patterns page, **Duplicate** shows the state of the mutate:

Duplicate	Meaning
Its normal light	No button held: **Duplicate** copies patterns, as before.
Dim green	**Shift** is held: a tap mutates.
Full green	**Shift** + **Duplicate** is down.
Green, pulsing	The Mutate settings page is open.
Red	**Clear** and **Shift** are held: a tap ends every mutate.

The settings

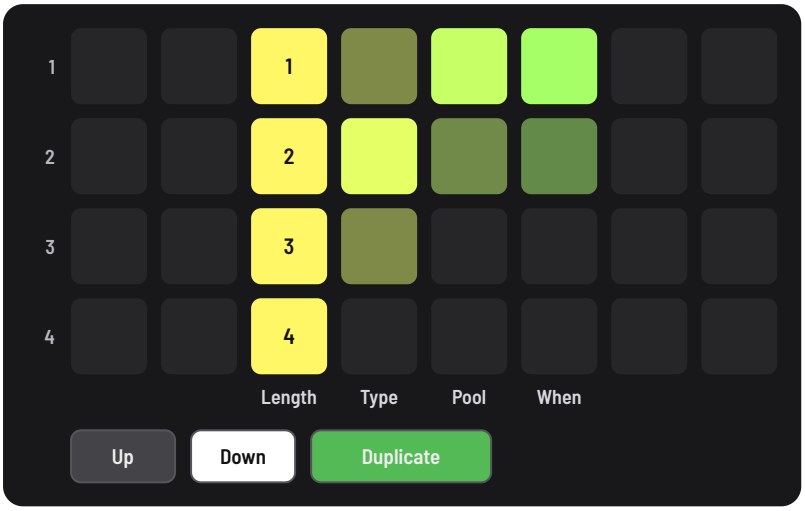
Hold **Shift** and hold **Duplicate** for a moment to open the Mutate settings page. Tap **Duplicate** to close it. The settings are for all tracks and are saved with the project. Chapters 2 and 3 show each setting.

> A track's mutate stays until you end it, pick a pattern for that track, or change the scene (see chapter 5).

CHAPTER 2

The Mutate settings, page 1

Hold Shift and hold Duplicate to open the page. Columns 3 to 6 are the settings; each pad in a column is one value. The chosen value is full, the other values are dim. Down shows page 2, Up shows page 1 again.



Page 1 with Length 4, Length type Exact, Pool With steps and When On the tap.

Column	Setting	Pads, top to bottom	What it does
3	Length	1, 2, 3, 4 (5-8 on page 2)	How many slots a list has, when Length type is Exact or Random. Tap the last pad of the bar.
4	Length type	Same as the chain (default) · Exact · Random	Same as the chain: a list as long as the track's chain (the Length column is very dim, not used). Exact: always Length slots. Random: 1 to Length slots, new at each mutate.
5	Pool	With steps (default) · All	With steps: only the patterns that have notes. All: all 8 patterns, also empty ones.
6	When	On the tap (default) · Every chain end	On the tap: the list starts at the next pattern end and loops. Every chain end: the track waits for its chain's end, then gets a new list each time its list ends.

The track buttons on this page

Tap a track button to choose if that track mutates. Bright = it mutates, very dim = it keeps its chain. All tracks are on at first.

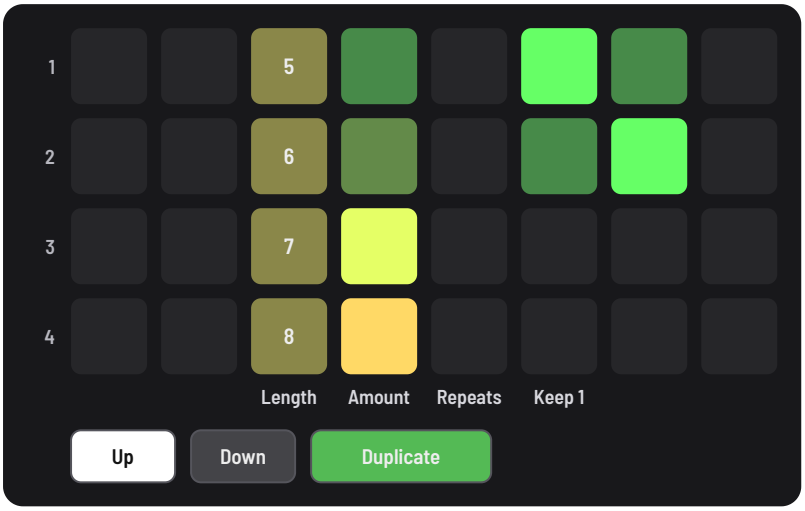
The Length bar

Length type	The bar in column 3
Same as the chain	Very dim: not used
Exact	Full up to the Length
Random	Dim, with the Length's pad full

CHAPTER 3

The Mutate settings, page 2

Press **Down** on the settings page. Column 3 goes on with Length 5-8. Columns 4 to 6 are three more settings. The greener colours tell you that you are on page 2.



Page 2 with Amount Some, Repeats On and Keep slot 1 On.

Column	Setting	Pads, top to bottom	What it does
3	Length	5, 6, 7, 8	The Length bar goes on from page 1.
4	Amount	All (default) · Most · Some · A little	How much of the chain a mutate changes. The meter fills from the bottom up: more light, more change.
5	Repeats	On (default) · Off	Off: a list never plays the same pattern twice in a row (also not from its last slot back to slot 1).
6	Keep slot 1	Off (default) · On	On: each list starts on the chain's first pattern, a home beat. Only the slots after it change.

Amount, step by step

Each slot of a list either keeps the chain's pattern at that place, or gets a random one. For a chain of 4 patterns:

Amount	Slots that change	Example: chain 1 · 2 · 3 · 4
All	4 of 4	7 · 6 · 2 · 5
Most	3 of 4	5 · 2 · 8 · 6
Some	2 of 4	1 · 6 · 3 · 2
A little	1 of 4	1 · 2 · 7 · 4

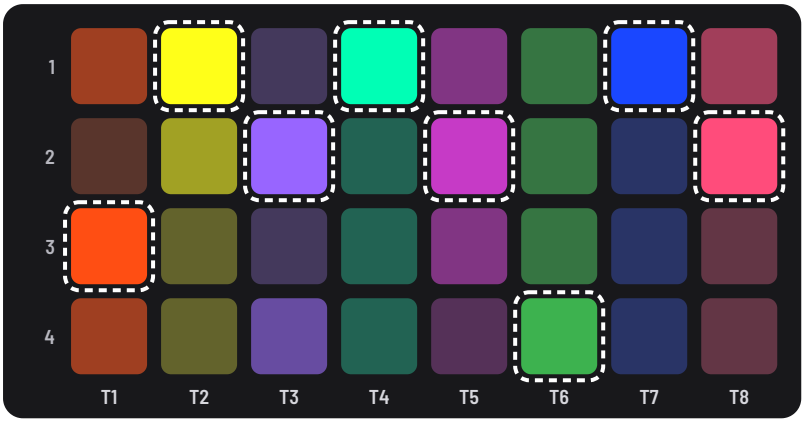
For other lengths the number is rounded, and at least one slot changes. Slots past the end of the chain are always random. With Keep slot 1 On, Amount counts only the slots after slot 1.

> With Repeats Off, a track that has only one pattern with notes still gets a list. It plays that pattern.

CHAPTER 4

Playing a mutate

On the Patterns page, each column is a track and each pad is a pattern. A mutated track shows its list:



Tracks 1, 3 and 6 are mutated. The other tracks show their chains as before.

In the list Not in the list (very dark) The pattern playing (it pulses)

You do	Result
Play	Each mutated track starts on slot 1 of its list.
Shift + Play	Play goes on from where you stopped. Then the list goes on from slot 1.
Mutate while stopped	The mutated tracks show their whole new list. The pad of the old stop point shows again when you play.
Tap a pattern pad of a mutated track	That track's mutate ends. The stock pattern pick plays, as before.
Clear + pad, Duplicate + pad	Clear or copy a pattern, as before. The mutate stays.
Shift held	The track buttons show their normal Shift lights. The pads keep showing the lists.

View Lock

Hold Shift and press Patterns to turn View Lock on or off. When you turn it on, the patterns that play at that moment turn white and stay white.

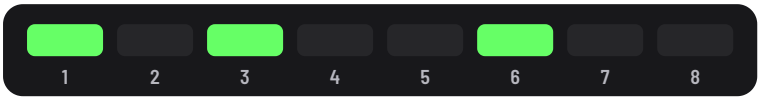
With View Lock on	Result
Shift + a pad	That track's white pad moves to this pattern. The mutate and the chain do not change.
A pad without Shift	The track's mutate ends and the chain is picked, as before. The white pads stay.
Clear + a green track button	The track's mutate ends (chapter 5). The white pads stay.
View Lock off, then on	The white pads move to the patterns that play now.

CHAPTER 5

Ending a mutate, scenes and saving

See which tracks are mutated

Hold Clear on the Patterns page. The track buttons of the mutated tracks are green; the other track buttons are dark.



Clear held: tracks 1, 3 and 6 are mutated.

You do	Result
Clear + a green track button	That track's mutate ends. Its button flashes red, then goes dark. The track goes back to its chain.
Clear + a dark track button	Nothing happens. The track is not selected.
Clear + Shift + Duplicate	Every mutate ends. Duplicate is red while Clear and Shift are held. The mutated tracks flash red.
Clear held on the settings page	The pads show the Patterns page, so you can see each mutate. Duplicate keeps pulsing green, so you know that you are still on the settings page. The pads do nothing. Clear + a track button ends that track's mutate.

On a track page, Clear + a track button is the track reset, as before.

Scenes

A scene keeps the mutate of each track, so a scene plays the exact lists that you heard when you saved it.

On the Mixer page	Result
Shift + a scene pad	Saves the scene with each track's mutate.
A scene pad	Plays the scene with its saved lists. Stopped: at once. Playing: when the scene starts.
Duplicate + a scene pad, then another	Copies the scene, its mutate too.
Clear + a scene pad	Clears the scene, its mutate too.

A scene saved without a mutate plays its chains: a scene change ends the mutate of the tracks with no list in that scene.

Saving

When you save the project, it keeps the Mutate settings, each track's current mutate and the mutate of each scene. A project from an earlier firmware opens with the default settings.

CHAPTER 6

Pattern Mutate quick reference

Control	Where	Result
Shift + tap **Duplicate**	Patterns page	Mutate (new lists with the settings)
Shift + hold **Duplicate**	Patterns page	Open the Mutate settings page
Tap **Duplicate**	Settings page	Close the settings page
Down / **Up**	Settings page	Page 2 / page 1
A setting's pad	Settings page	Choose that value
A track button	Settings page	That track mutates or not
Clear held	Settings page	Show the mutate; pads do nothing
Clear + a track button	Settings page	End that track's mutate
Clear held	Patterns page	Mutated tracks' buttons green
Clear + a green track button	Patterns page	End that track's mutate
Clear + **Shift** + **Duplicate**	Patterns page	End every mutate
A pattern pad	Patterns page	End that track's mutate, pick the pattern
Shift + a pad (View Lock on)	Patterns page	Move the white pad; the mutate stays
Play	Any page	Start the lists from slot 1
Shift + **Play**	Any page	Go on from the stop point
Shift + a scene pad	Mixer page	Save the scene with the mutate

Defaults

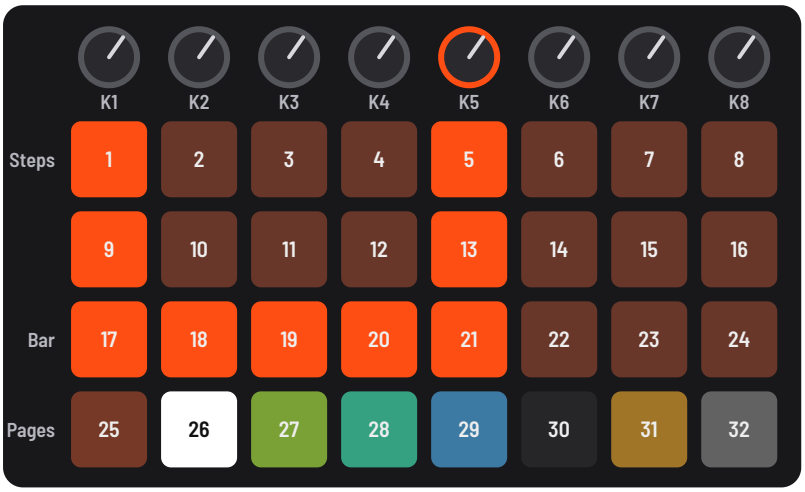
Setting	Default
Length	4
Length type	Same as the chain
Pool	With steps
When	On the tap
Amount	All
Repeats	On
Keep slot 1	Off
Tracks	All on

CHAPTER 7

Track +: overview and layout

Track + is the track menu. It shows the eight knobs of the selected track on pages: the stock's own parameters and the extra parameters that Rhythm Plus adds, such as envelopes, LFOs, movement, glide and ring modulation.

1. Select a track. Tap its track button again to open Track +.
2. Tap a page pad (26-31) to choose a page. Each track keeps its own page.
3. Turn a knob. The bar shows its value. Tap the track button again to close Track +.



Track + on track 1, the Parameter Editor (pad 26) open, K5 shown on the bar.

Pads	Name	What they do
1-16	Steps	The steps of the pattern, as on the track page. Hold a step to view or lock its values.
17-24	Bar	The value of the shown knob.
25	Knob pick	Shows the page's next knob on the bar, without turning it.
26	Parameter Editor (white)	The stock's eight knobs of the track.
27-31	Extra pages	27 Filter (lime), 28 Mix (mint), 29 Pitch (sky blue), 30 none (dark), 31 FX (amber).
32	All	With **Clear** held: resets all pages (chapter 10).

Terms

Term	Meaning
Shown knob	The last knob that you turned, or the knob that you picked with pad 25. Its light is in the track's colour.
Edit	Turn the knob to change the value for the whole track.
Automate	Arm Record and turn the knob while the pattern plays. The movement is recorded into the steps.
Lock	Hold a step, arm Record and turn the knob. The value goes into that step only.
Reset	Hold **Clear** and turn the knob right until it stops. The value goes back to its default; the knob light turns blue.
Delete automation	Hold **Clear** and turn the knob left until it stops. The knob's automation is deleted; the knob light turns red.

CHAPTER 8

The Parameter Editor (pad 26)

These are the stock's parameters of the track. You can view, edit, automate, lock, reset and delete the automation of all eight.

Knob	Parameter	Shift + knob	Bar
K1	Pitch	-	From the middle
K2	Sample start	Fine start	2048 steps a pad
K3	Sample length	Fine length	2048 steps a pad
K4	Envelope: left of the middle = decay, middle = sustain, right = slow attack	-	From the middle
K5	Distortion	-	From the left
K6	High-pass filter	-	From the left
K7	Low-pass filter	-	From the left
K8	Resonance	-	From the left

View a step's value

Hold a step on pads 1-16. Each knob light turns red if that knob has automation on the step. If the shown knob has automation on the held step, the bar shows the step's value.

Automate and lock

To	Do this
Automate	Arm Record, play, turn the knob. The movement is recorded into the steps that play.
Lock	Arm Record, hold a step, turn the knob. The value goes into the held step, also while the pattern plays.

While Record is armed and you turn a knob, the bar is red. While you hold a step with Record armed, the step's pad is red.

CHAPTER 9

The extra pages (pads 27-31)

The extra parameters are kept for each track and saved with the project. You can view, edit and reset them; they have no automation and no step locks. Filter K7 and K8 are the stock's low-pass and resonance (white lights): you can automate and lock them, as on the Parameter Editor.

Shift on these pages: on a shape knob, or a rate knob in its synced half, Shift + turn right goes to the same pad on the second sweep, Shift + turn left goes back. On the other knobs, Shift + turn moves the value in fine steps.

Filter (pad 27, lime)

Knob	Parameter	Shift + knob
K1	Envelope amount (from the middle)	Fine
K2	Envelope shape	2nd sweep
K3	Envelope decay	Fine
K4	LFO depth	Fine
K5	LFO shape	2nd sweep
K6	LFO rate: right = synced, left = free	2nd sweep / fine
K7	Low-pass (stock)	Filter type: stock or 24 dB
K8	Resonance (stock)	Pitch follow

Mix (pad 28, mint)

Knob	Parameter	Shift + knob
K1	Volume movement depth	Fine
K2	Volume movement shape (10 LFO, 6 gates)	2nd sweep
K3	Volume movement rate: right = synced, left = free	2nd sweep / fine
K4	Pan movement depth	Fine
K5	Move shape (pan and sends)	2nd sweep
K6	Move rate: right = synced, left = free, middle = off	2nd sweep / fine
K7	Reverb send movement (from the middle)	Fine
K8	Delay send movement (from the middle)	Fine

The extra pages (pads 27-31)

Pitch (pad 29, sky blue)

Knob	Parameter	Shift + knob
K1	Envelope amount (from the middle)	Fine
K2	Envelope shape	2nd sweep
K3	Envelope decay	Fine
K4	LFO depth (vibrato)	Fine
K5	LFO shape	2nd sweep
K6	LFO rate: right = synced, left = free	2nd sweep / fine
K7	Random pitch, up to 12 semitones	LFO fade-in, up to 2 s
K8	Glide, up to about 1 s	Fine

FX (pad 31, amber)

Knob	Parameter	Shift + knob
K1	Ring modulation amount	Fine
K2	Ring modulation frequency	Fine
K3-K6	None (dark)	-
K7	Scan depth: moves the sample start	Fine
K8	Scan speed: right = synced, left = free	2nd sweep / fine

Pad 30 has no page; it is dark.

CHAPTER 10

Resetting values and pages

One knob

Do this	Result
Clear + turn the knob right until it stops	The value goes back to its default. The knob light turns blue.
Clear + turn the knob left until it stops	Parameter Editor (and Filter K7 / K8): the knob's automation is deleted; the light turns red.

To reset the Pitch page's LFO fade-in (Shift + K7): hold Clear, then hold Shift, then turn K7 right. Press Clear first: Shift + Clear changes the metronome.

A whole page

Do this	Result
Clear + tap pad 26	The eight stock parameters go back to their defaults, and their automation in the current pattern is deleted.
Clear + tap a page pad (27-31)	All extra parameters of that page go back to their defaults.
Clear + tap pad 32	All of the above: the stock parameters, their automation in the current pattern, and the extra parameters of all pages.

Page resets work on the selected track only. They do not change the sample, the mixer or the FX sends. The Filter page's K7 and K8 (the stock's low-pass and resonance) change only with pad 26 and pad 32, not with pad 27.